Transformers Find the Item but Misread It: Locating versus Readout in Working Memory

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Abstract

Large language models fail the N-back task in a human-like way: accuracy falls steeply as N grows, and most errors are false alarms on near-target lures. A prominent account attributes this limit to the precision of softmax attention, which must single out one past position among many. We test that account directly in 15 pretrained models (Pythia 70M–6.9B, GPT-2 124M–1.5B, Qwen2.5 0.5B–14B) and in minimal transformers trained from scratch. We separate each error into a *locating* failure (attention lands on the wrong item) and a *readout* failure (attention lands on the right item but the answer is still wrong). Three results favour readout. First, the heads that carry the task attend to the target on about 90% of misses. Second, linear probes on the residual stream separate true matches from lures almost perfectly (median AUC 0.98–1.00) while the model's own answer barely does (0.58–0.63), including on the model's own false alarms. Third, sharpening the identified heads barely changes performance (median $\Delta d' = +0.05$), whereas removing the letter identity of non-target items raises it in 42 of 43 model×N cells (median $+0.60$, up to $+1.78$ in Qwen2.5-14B), five times the gain of a recent content-suppression intervention we replicate. In minimal models, softmax precision does limit a one-layer, residual-free transformer, but depth and a residual stream remove the limit entirely, and even there accuracy tracks how content is used rather than how sharply position is coded. Working-memory limits in transformers are therefore limits of readout under interference, not of attention precision.

1 Introduction

Working memory, the ability to hold and use a few items over seconds, has a famously small capacity in humans [Cowan, 2001, Ma et al., 2014]. The N-back task probes it directly: for each item in a stream, say whether it matches the one N steps back [Kirchner, 1958]. People get worse as N grows, and their errors concentrate on lures, items that match at N±1 [Kane et al., 2007]. Executive-attention accounts explain this as a failure to select the target against competitors [Engle, 2002].

Pretrained language models show the same signature even though the whole sequence sits in their context [Gong et al., 2024, Hu and Lewis, 2025]. Because nothing has been forgotten, the bottleneck must lie in *retrieval*. Two mechanistic accounts compete. The *precision* account holds that softmax attention cannot concentrate enough mass on position $i - N$ when many keys compete, so nearby items leak in [Gong and Zhang, 2024, Veličković et al., 2025]. The *interference* account holds that the model can find the target, but the content of other items is entangled with it in the residual stream and corrupts the decision [Xiong et al., 2026]. The two predict different things: if attention is imprecise, errors should come with attention in the wrong place and sharpening attention should help; if readout is the bottleneck, errors should occur with attention in place, the needed information should be present in the representation, and removing competing content should help.

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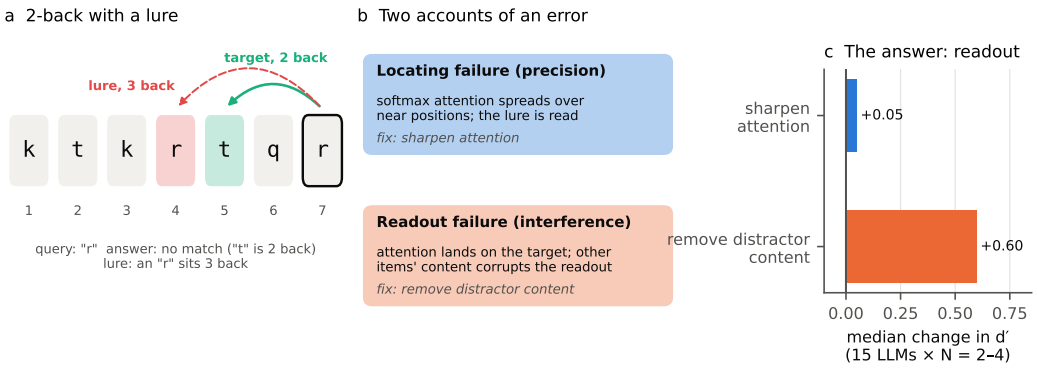


Figure 1: **Overview.** (a) N-back with a lure: the target is the item N back; a lure repeats the query letter at N±1. (b) Two accounts of an error. (c) Across 15 LLMs and N = 2–4, sharpening the heads that carry the task barely changes d', while removing the letter identity of non-target items raises it substantially (medians; Section 7).

We test both predictions with a common set of stimuli, thresholds and interventions across 15 pretrained models and a family of minimal transformers. Our contributions:

- **A locating-versus-readout decomposition** (Section 6). Content-addressed heads place attention on the target line on about 90% of misses; lure false alarms come from heads landing on the same-letter lure. Probes show that whether a match sits at $i - N$ or at a lure offset is linearly decodable in the residual stream, and remains so on the model's own false alarms.
- **A causal contrast on one axis** (Section 7, Fig. 5). Ablating the identified heads costs a median 0.93 d'; sharpening them gains a median 0.05; removing distractor content gains 0.60, with sequence-level bootstrap intervals for every cell.
- **Where precision binds** (Section 5). We show when the softmax precision bound constrains capacity (one layer, no residual stream) and why depth removes it. We prove that in this model content-independent attention carries no signal however precise it is, and give a readout model that separates sharpening from removing interference. Each prediction is tested (Table 1); accuracy tracks the use of content, not the sharpness of position.
- **Architecture and development** (Section 8). Fixed-state recurrent models with enough state solve the task, while a selective state-space model declines with N like attention; N-back heads appear early in Pythia training.

2 Related work

Working memory in LLMs. Gong et al. [2024] measured N-back capacity in ChatGPT and found human-like declines; Gong and Zhang [2024] trained small transformers on N-back and related the decline to attention entropy. Hu and Lewis [2025] showed that weaker models drift to an easier task, a confound we control by reporting the best-fitting k for every cell. Zhang et al. [2024] relate working-memory tests to reasoning limits. Closest to us, Xiong et al. [2026] show that LLMs fail working-memory tasks that a two-layer transformer solves, attribute the failure to interference among entangled item representations, and report gains from suppressing letter-identity directions, selected as the best configuration of a sweep on the evaluation trials. We replicate their intervention with held-out selection and compare it directly with head sharpening and with targeted distractor suppression.

Softmax precision. Softmax attention disperses as the number of keys grows [Veličković et al., 2025, Hayase and Karakida, 2026]; scalable softmax [Nakanishi, 2025] and sigmoid attention [Ramapuram et al., 2025] change this behaviour. Expressivity results bound what one attention layer can compute [Hahn, 2020, Sanford et al., 2023]. We use these ideas to state the N-back-specific bound and test whether it is the binding constraint.

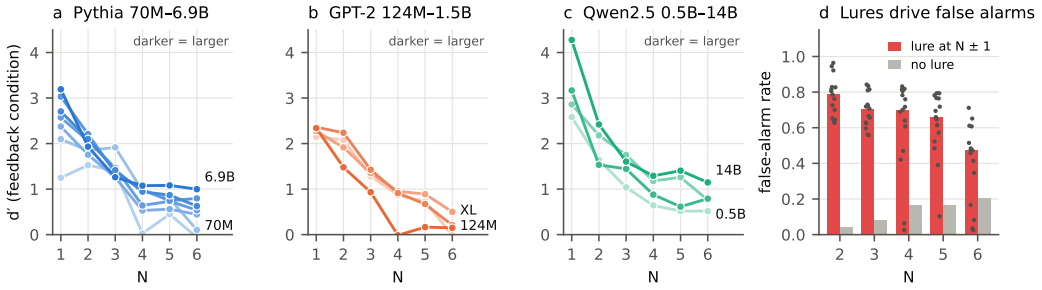


Figure 2: **Behaviour.** (a–c) d' versus N for three model families; darker lines are larger models. (d) False-alarm rates on lures at N$\pm$1 (red; dots are models) and on non-lure items (gray), medians over 15 models.

Mechanistic interpretability. We build on circuit analyses [Elhage et al., 2021, Wang et al., 2023], induction heads [Olsson et al., 2022], retrieval heads [Wu et al., 2025], and activation and attribution patching [Zhang and Nanda, 2024, Kramár et al., 2024]. Recurrent and state-space baselines follow Hochreiter and Schmidhuber [1997], Orvieto et al. [2023], Gu and Dao [2024], Katharopoulos et al. [2020]; associative recall in such models is studied by Arora et al. [2024], Jelassi et al. [2024].

3 Setup

Task and stimuli. Sequences of 24 letters from a 20-consonant alphabet, with match rate fixed per N and four planted lures per test sequence (a letter repeating the query at N$\pm$1 or another non-target offset, never creating a match). LLM prompts show three solved demonstration sequences, then the test sequence one item per line ("`k -`", "`t m`"), with correct answers shown for earlier items (*feedback* condition). The score at item i is $\log p(\texttt{m}) - \log p(\texttt{-})$ at the letter token. Thresholds are calibrated on 200 held-out sequences per model, N and condition; all metrics use 1,000 test sequences. A chat format following Gong et al. [2024] is used for Qwen2.5-Instruct models (Appendix B).

Measures. Sensitivity d' with the log-linear correction [Hautus, 1995], hit and false-alarm (FA) rates, FA rates split by lure offset, and the best-fitting k (the k-back rule the model's answers fit best) to detect task drift.

Models. Pythia 70M–6.9B [Biderman et al., 2023], GPT-2 124M–1.5B [Radford et al., 2019] and Qwen2.5 0.5B–14B [Qwen Team, 2024]. Minimal models are attention-only transformers trained from scratch to convergence, varied in depth, residual stream, position encoding and attention normalization (Appendix B).

Heads. For each model and N we rank heads by activation patching from a clean to a corrupted run in which the correct answer flips, scoring the fraction of the mean clean–corrupt logit gap that patching one head recovers [Zhang and Nanda, 2024]. One head typically dominates and is shared across N; at N = 2 the top head recovers 16–73% of the gap.

4 Behaviour: the decline and its lures

All 15 models decline steeply with N (Fig. 2a–c; Table 2). Median d' falls from 2.57 at N = 1 (range 1.25–4.28) to 0.91 at N = 4 and 0.52 at N = 6. Scale helps at N = 1 but little beyond: Qwen2.5-14B reaches 4.28 at N = 1 and 1.15 at N = 6. Errors are dominated by lures (Fig. 2d): at N = 2 the median FA rate is 0.79 on lures at N$\pm$1 against 0.04 on non-lure items. Task drift appears only at large N in small models (best-fitting $k < N$ in 14 of 90 base-model cells, all at N$\geq$4). The chat format reproduces the decline in Qwen2.5-Instruct 0.5B–7B (d' from 1.8–2.0 at N = 1 to 0.4–0.7 at N = 6; Appendix C).

5 When does attention precision bind?

Consider the minimal model of Gong and Zhang [2024]: one attention layer, no residual stream and no MLP, with output $o_i = \sum_j a_{ij} v_j$, a linear decision on o_i, and logits $s_{ij} = c(x_i, x_j) + p(i-j)$ (a content and a position term). For a query at position i, let $t = i - N$ be the target, $u_j = w^\top v_j$ the scalar the readout assigns to key j, and $r_i = \sum_j a_{ij} u_j$ the decision variable. Without a residual stream, $u_j = u(x_j, j)$ depends only on the letter and position of key j. A trial is a match when $x_t = x_i$. The stimulus generator treats all letters alike (it is invariant under relabelling the alphabet), so each single position has the same, uniform letter distribution on match and non-match trials; only joint distributions differ.

Proposition 1 (the readout is a mass test)**.** *In this model the decision is linear in $\sum_j a_{ij} v_j$, and for fixed weights a_{ij} no linear function of a convex combination of letter embeddings can test whether two letters are equal. The difference in mean readout between match and non-match trials must therefore be carried by how the attention distribution differs between them.*

Corollary 1 (precision without content carries no signal)**.** *For any fixed weight vector $\bar{a}$ and $u_{\max} = \max_{x,j} |u(x,j)|$,*

$$\big|\mathbb{E}[r_i \mid \textit{match}] - \mathbb{E}[r_i \mid \textit{non-match}]\big| \;\le\; u_{\max}\big(\mathbb{E}_{\textit{match}}\|a_i - \bar{a}\|_1 + \mathbb{E}_{\textit{non-match}}\|a_i - \bar{a}\|_1\big).$$

In particular, any attention pattern that does not depend on content, including an oracle that splits all mass between i and $i-N$, gives the same mean readout on match and non-match trials, however precisely it locates the target; only differences in the shape of the two distributions remain for a threshold to use.

Proposition 2 (softmax dispersion)**.** *If each of the K other keys has a logit at least Δ below the target's, the target receives mass at least $1/(1 + Ke^{-\Delta})$, and this guarantees mass $1-\epsilon$ for every such configuration if and only if $\Delta \ge \log(K(1-\epsilon)/\epsilon)$. Two keys with the same letter are separated only by the position term: a key at offset $N \pm 1$ receives $e^{-\delta_P}$ times the mass of a key at offset N, with $\delta_P = p(N) - p(N \pm 1)$.*

Corollary 2 (what the precision account predicts)**.** *Suppose the decision is driven by the mass on a key that carries the query letter (the attend-on-match solution). Compare a match trial with a lure trial that is identical except that the query letter sits at offset $N \pm 1$ instead of N. The lure key then receives $e^{-\delta_P} Z^m / Z^l$ times the mass the target receives on the match trial, where Z^m, Z^l are the two softmax normalizers. Unless the normalizer ratio compensates, lure false alarms should fall as δ_P grows, and if dispersion caused the decline with N, δ_P should shrink as N grows.*

Proposition 3 (depth removes the bound)**.** *With two layers, a residual stream and an absolute or rotary position code, a first-layer head can copy x_{i-N} into position i with margin growing in the weight norm. Position i then carries both letters, and a second layer (through a bilinear query–key term) or an MLP can test their equality without competing against the other, N-dependent keys. A monotone recency bias (ALiBi, or no position code) cannot single out offset $N > 1$ in one head, so the copy step fails.*

Once depth removes the bound, what limits a deep model? Proposition 4 describes a general linear readout in which other items' content interferes with the decision. It does not assume one layer, and it separates the two interventions of Section 7.

Proposition 4 (sharpening versus removing interference)**.** *Let the decision variable be $r = w^\top h$ with $h = \gamma\, y\, e + \sum_{j \in D} B_j g_j + z$, where $y = \pm 1$ marks match and non-match, γe is the signal, $g_j \in \mathbb{R}^d$ is the letter content of distractor item $j \in D$, which lies in a q-dimensional subspace, B_j are fixed linear maps that write it into h, and z is noise independent of the letters. Distractor content may depend on y (lures are planted on non-match trials): let $b_y = \mathbb{E}[w^\top \sum_j B_j g_j \mid y]$, $\beta = b_+ - b_-$, and let σ_D^2 and σ_z^2 be the within-class variances of the distractor and noise terms of r. Then $d' = (2\gamma\, w^\top e + \beta)/\sqrt{\sigma_D^2 + \sigma_z^2}$.*

- *(a) Removing a fraction α of every distractor's letter content ($g_j \to (1-\alpha) g_j$) gives $d'(\alpha) = (2\gamma\, w^\top e + (1-\alpha)\beta)/\sqrt{(1-\alpha)^2 \sigma_D^2 + \sigma_z^2}$, which increases with α when $\beta \le 0$, that is, when distractors push non-match trials towards "match", as lures do.*
- *(b) Removing a uniformly random q-dimensional subspace instead removes, in expectation, a fraction q/d of each distractor's content (in squared norm).*

Table 1: Predictions of the propositions and their tests. ✓: confirmed; ×: not supported.

Statement	Prediction	Test	Result
Cor. 1	Content-independent attention gives no mean signal, however precise	E7: fixed mix of i, $i-N$; content term removed	$\lvert d'\rvert \leq 0.07$ at every N ✓
Prop. 1	d' tracks the match/non-match attention-mass gap	300 one-layer checkpoints	$\rho = 0.62$ (Fig. 3e) ✓
Prop. 2	Changing the normalization changes the decline	E5: sigmoid, top-k, scalable softmax	sigmoid and top-k flatten it ✓
Cor. 2	Lure FAs fall with δ_P; δ_P shrinks with N	E1 × E6 logit decomposition	$\rho = -0.03$; δ_P 4.5 → 6.7 ×
Prop. 3	Depth and residual remove the decline with absolute or rotary codes	E3b, E10	d' 4.5–7.8, flat ✓; ALiBi, none fail (1.3–1.8)
Prop. 4	Removing distractor content beats sharpening, cuts FAs, keeps hits; random subspace ≈ 0	B1 in 15 LLMs (Sec. 7)	$+0.60$ vs. $+0.05$; random -0.03 ✓

(c) If a head with value vectors v_j and output $W_O \sum_j a_j v_j$ reaches h directly and puts mass at least $1-\epsilon$ on the target line, it contributes at most $\epsilon \max_j |w^\top W_O v_j|$ of distractor content to r; this bounds what re-weighting its attention can remove.

Sharpening thus acts on one head and can remove at most that head's off-target share of interference; when off-target mass carries signal, as in the minimal model (Corollary 1), it removes signal as well. Removing distractor content acts on interference wherever it is written, by any head or layer, and removes both the lure bias β and the variance σ_D^2: false alarms should fall while hits hold. When the top heads carry a small share of the interference, suppression should therefore beat sharpening. The linear model predicts gains that grow with α. In LLMs the gain peaks at $\alpha = 0.5$ (Section 7), so full removal also costs something, for instance by moving activations off their usual range. Proofs of all statements are in Appendix A; Table 1 lists each prediction and its test.

Tests in the minimal model. Figure 3 tests Propositions 1–3. The one-layer model declines with N (d' 2.67 to 2.15 over 50 seeds; panel a). Unnormalized sigmoid attention and top-k attention raise and flatten the curve, while scalable softmax does not (panel b), as Proposition 2 predicts for a normalization-driven limit. With two layers and a residual stream the decline disappears: d' is 6.4–7.2 at every N with learned positions, 5.4–5.8 with sinusoidal and 4.5–5.9 with rotary codes, and 7.7–7.8 for a single model trained on all N with a cue (panel c); ALiBi and no-position models fail at every N (d' 1.3–1.8). Proposition 3 rules out their copy step for $N > 1$; their failure at N = 1, where the most recent item is the target, is stronger than it predicts.

Corollary 1 is confirmed. On 720 trained one-layer checkpoints, fixing attention to any mix of i and $i-N$, which keeps both letters in the output, gives $|d'| \leq 0.07$ at every N, and removing the content term of the logits gives $|d'| \leq 0.04$. Removing the position term also fails (d' between -0.28 and -0.03), because content alone cannot tell offset N from any other repeat. Precise location is useless without content-dependent attention.

Is dispersion the binding constraint even in the one-layer model? The data reject both predictions of Corollary 2. Decomposing the logits of the 720 checkpoints, the position margin δ_P does not shrink as N grows (4.5 at N = 1, 6.7 at N = 5). Across the 300 checkpoints of the main configuration, δ_P predicts neither lure false alarms (Spearman $\rho = -0.03$) nor d' ($\rho = -0.05$; Fig. 3d). What does predict d' is the size of the difference in target attention between match and non-match trials ($\rho = 0.62$; leave-one-N-out $R^2 = 0.37$; panel e), as Proposition 1 implies. The sign of that difference changes with N: every seed attends more to the target on match trials at N = 1, but only 48% do at N = 4 and 18% at N = 6 (panel f); the rest attend away from a matching letter. Both solutions are mass tests, and Corollary 1 bounds them alike. Sharpening the trained attention also lowers d' at N = 1–5 (2.70 to 2.20 at N = 1 with $\tau = 0.67$; at most $+0.06$ at N = 6), and removing off-target mass costs more (Appendix C): off-target mass is part of the code. Precision bounds the minimal model, but the decline is shaped by how content is read out.

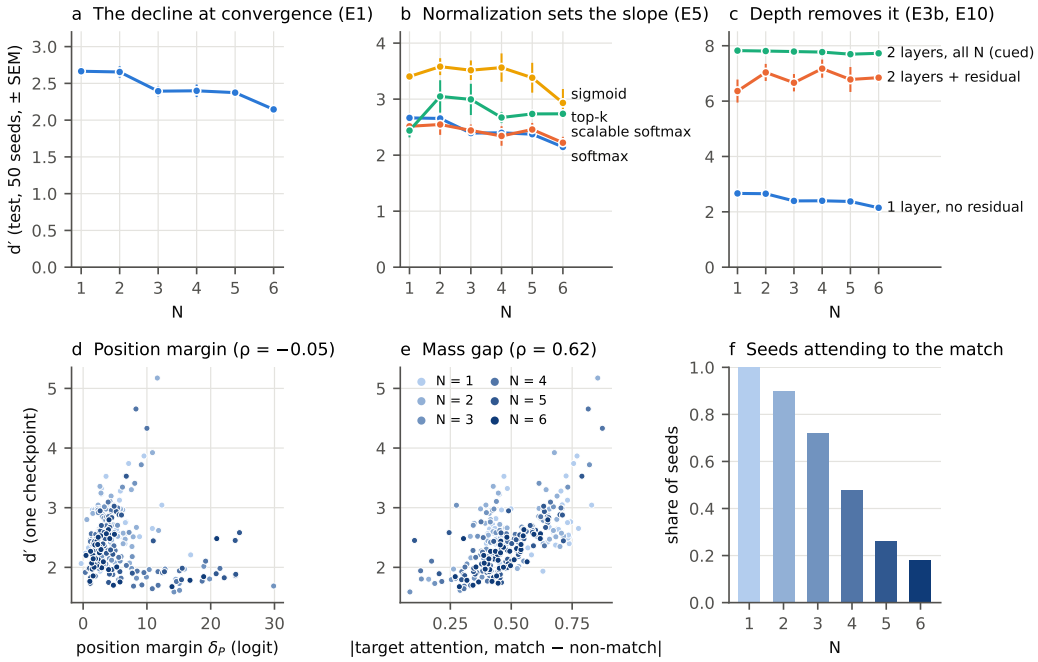


Figure 3: **Minimal models.** (a) The one-layer, residual-free model declines with N at convergence (50 seeds, ±SEM). (b) Sigmoid and top-k attention flatten the decline; scalable softmax does not. (c) Two layers with a residual stream remove it, for one N per model or all N in one cued model. (d, e) Across 300 one-layer checkpoints (50 seeds × N = 1–6; darker = larger N), d' is unrelated to the position margin δ_P but tracks the absolute difference in target attention between match and non-match trials. (f) Share of seeds whose target attention is higher on match trials.

6 Locating versus readout in LLMs

Do errors come with attention in the wrong place? For every test item we measure the attention of the top-5 heads from the query letter to the whole line (letter, answer, newline) of item $i - N$ and of its neighbours $i - N \pm 1$, and call the item *located* when the target line receives at least as much attention as either neighbour. On correct matches the heads locate the target in a median 88–91% of items; on correct non-matches only about 25%, because nothing at $i - N$ matches: the heads are content-addressed. On misses the target is located just as often: attention is elsewhere on only 16%, 7% and 8% of misses at N = 2, 3 and 4 (Fig. 4a; medians over 14 models, excluding GPT-2 Large, whose top heads attend to none of the three lines). A logistic probe trained on correct trials then reads the answer from the heads' concatenated outputs: the heads' output already pointed to the wrong answer in 41–57% of misses, and to the right answer, lost in later layers, in 27–43%. Lure false alarms are the mirror image: on 91–96% of them the target line loses to a neighbouring line, typically the lure's, which holds the matching letter.

Is the position of the match represented? Matches and lures both contain the query letter, so only its position separates them. A logistic probe (5-fold, split by sequence) reads match versus lure from the query token's residual stream at nine depths (Fig. 4b,c). The model's own score separates them with median AUC 0.63 (N = 2) and 0.58 (N = 3). The probe reaches a median of 1.00 and 0.98 at the best depth; the input embedding is at chance. On the model's own lure false alarms, the probe still calls a median 95% (N = 2) and 86% (N = 3) of items lures. The largest model shows the same gap: in Qwen2.5-14B the model's own score separates matches from lures with AUC 0.81 and 0.64, the probe with 1.00 and 0.997, and the probe calls 99.8% and 95% of the model's lure false alarms lures. The information peaks in middle layers and declines toward the output (N = 3: 0.92 at the last layer), and the top heads' outputs carry less of it (AUC 0.71–1.00) than the residual stream. Only in the GPT-2 family at N = 3 is part of the information lost (probe AUC 0.84–0.89). The models compute where the match was, then do not use it.

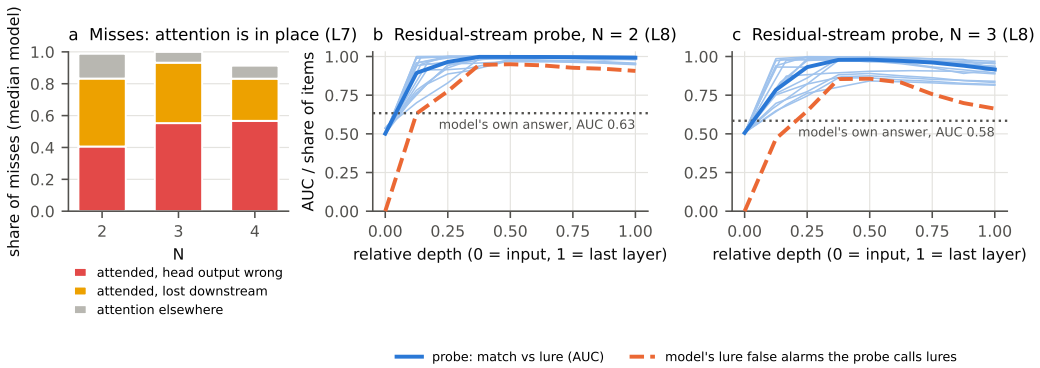


Figure 4: **Locating versus readout.** (a) Misses split by whether the top heads attended to the target line and whether their output carried the right answer (median model). (b, c) A probe separates matches from lures by the position of the matching letter (blue; thin lines are models) and recognizes most of the model's own lure false alarms (orange), far above the model's own answer (dotted).

7 Causal tests: sharpening versus removing interference

We apply five interventions to all 15 models at N = 2–4, on the same sequences, each with its own calibrated threshold (Fig. 5; Table 3). Intervals are 95% paired bootstrap intervals over sequences (2,000 resamples); two degenerate cells whose baseline d' is below 0.3 are excluded.

- **Ablation.** Mean-ablating the top-10 patching heads lowers d' in 42 of 43 cells (median -0.93); ten random heads from the same layers do not (median $+0.00$). The effect shrinks with scale (Qwen2.5-7B: -0.27 to $+0.03$; Qwen2.5-14B: -0.48 to -0.17), indicating more distributed solutions.
- **Sharpening.** Scaling the top-3 heads' logits by $1/\tau$ changes d' by a median $+0.05$ at $\tau = 0.67$ and $+0.03$ at $\tau = 0.8$ (range -0.30 to $+0.30$); flattening them ($\tau = 2$) costs a median 0.32 in the full sweep. Changing the temperature of every head helps most at mild *flattening* ($\tau = 1.25$) in 19 of 30 cells. The heads sit near their optimal sharpness.
- **Distractor suppression.** For each query i, we remove the 19-dimensional letter-identity subspace (fit on calibration sequences at every residual site) from the whole line of every earlier item except $i - N$, scoring each query on its own prefix. At half strength this raises d' in 42 of 43 cells (median $+0.60$, up to $+1.78$), including at N = 4 (median $+0.44$) and in Qwen2.5-14B ($1.60 \rightarrow 3.38$ at N = 3). False alarms fall from about 0.2–0.3 to about 0.1 while hits hold. Removing a random subspace of the same rank does nothing (median -0.03), as Proposition 4b predicts ($q/d \leq 19/512$); removing the target's letter destroys performance (median -1.54). Full strength also helps (median $+0.46$) but over-suppresses in Qwen2.5-7B.
- **Replication of Xiong et al. [2026].** One letter-identity direction at one of the two earliest depths, applied at answer positions only, with the configuration chosen on calibration sequences: median $+0.11$ (range $+0.01$ to $+0.30$, 28 cells). The best-of-sweep statistic selected on test data gives $+0.12$, so their effect replicates; it is about a fifth of the effect of targeting the distractor items.

Across cells, distractor suppression and head ablation differ from zero with Wilcoxon $p < 10^{-8}$; sharpening gives a small but consistent gain ($p = 0.04$ at $\tau = 0.67$, $p = 0.001$ at $\tau = 0.8$), an order of magnitude below distractor suppression. Proposition 4 explains this ordering: sharpening can remove only the top heads' off-target share of interference, while suppression removes it, and the lure bias with it, wherever it is written. Distractor suppression is oracle-targeted, since it knows which item is the target, so it is a causal test of interference rather than a usable method.

Scale. The two accounts part ways as models grow (Fig. 5b). Ablating the identified heads costs less in larger models (Spearman $\rho = 0.74$ between size and each model's median change, $p = 0.002$, 15 models), from about -1.6 in the smallest Pythia models to -0.21 in Qwen2.5-14B: the solution

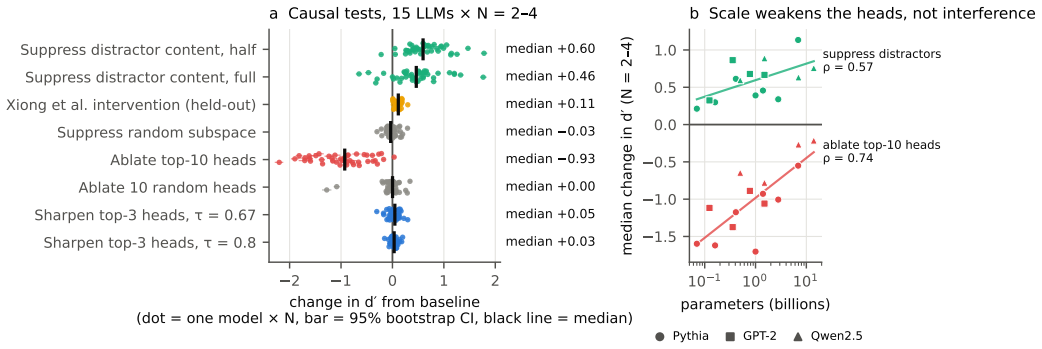


Figure 5: **Causal tests.** (a) Change in d' from baseline for each intervention; each dot is one model $\times$ N cell with its 95% bootstrap interval, black lines are medians. (b) Each model's median change over N = 2–4 against its size, for distractor suppression (half strength) and top-10 head ablation; lines are least-squares fits in log size.

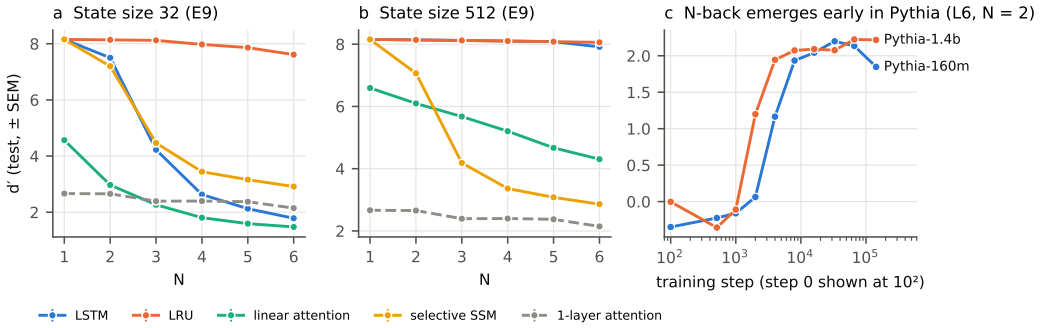


Figure 6: **Architectures and development.** (a, b) Recurrent and linear-attention baselines at state sizes 32 and 512, with the one-layer attention model (dashed) for reference; LSTM-512 overlaps LRU-512. (c) N-back d' over Pythia training (N = 2).

becomes more distributed. Removing distractor content helps more ($\rho = 0.57$, $p = 0.03$, 15 models). Qwen2.5-14B, the largest model, shows the pattern most clearly. Half-strength suppression raises its d' from 2.42 to 2.88, 1.60 to 3.38 and 1.29 to 2.05 at N = 2, 3 and 4, with every bootstrap interval at least 0.40 above zero; a random subspace of the same rank changes it by -0.10, -0.05 and $+0.18$. With distractor content removed, the model at N = 3 outperforms its own unmodified N = 2 baseline. Scale makes the task less dependent on a few heads, but not less limited by interference.

8 Architectures and development

Recurrent baselines trained on the same stimuli separate two kinds of memory (Fig. 6a,b). LSTM and LRU with a 512-dimensional state solve every N ($d' \geq 7.9$); with a 32-dimensional state the LSTM declines (8.2 to 1.8) while the LRU barely does. Linear attention declines at both sizes, and a selective state-space model, whose gating is input-dependent like attention, declines from 8.2 to 2.9 even with a 512-dimensional state. The decline is not specific to softmax attention; it appears in architectures that retrieve by content-dependent weighting. In Pythia, the top N-back heads and the behaviour appear early in training: d' at N = 2 rises from chance to near its final value between steps 1,000 and 8,000, alongside previous-token and induction heads (Fig. 6c).

9 Discussion and limitations

Pretrained transformers locate the item N back: content-addressed heads land on the target line, and the residual stream encodes whether the matching letter sat at $i - N$ or at a lure position. They

fail in readout, where the content of other items interferes with the decision. Sharpening attention cannot fix this; removing the interfering content largely does. The precision account is not wrong so much as narrow: it describes a one-layer, residual-free model, and depth removes it. This places transformer working memory closer to interference accounts of human working memory than to fixed-slot or attention-bottleneck accounts [Ma et al., 2014, Engle, 2002], and suggests that training or architectural changes that separate item content from the decision variable, rather than sharper attention, are the lever for improving it.

Limitations. We study one task family (letter N-back) with synthetic stimuli. The mechanistic analyses use the line format and the feedback condition. Models range up to 14B parameters, and the identified heads matter less in larger models, so the head-level account may not transfer to frontier models, although the suppression effect grows rather than shrinks with scale in our range (Fig. 5b). Distractor suppression requires knowing the target and is a diagnostic, not a method. The probes show that information is linearly present, not that the model could use it with its existing weights. Propositions 1–3 concern the minimal model; Proposition 4 is a linear model of the readout, not derived from trained weights, and does not explain why suppression helps most at half strength.

Broader impact. This is basic research on model mechanisms with public models and synthetic data. Better diagnosis of working-memory failures may help evaluation; the interventions are analysis tools with no deployment use.

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A Proofs

Proposition 1. Let e_x be the value vector of letter x (no residual stream, no MLP, so the output is $o_i = \sum_j a_{ij} e_{x_j}$ up to a fixed linear map, plus positional components that do not depend on the letters). A linear readout computes $w^\top o_i = \sum_j a_{ij} w^\top e_{x_j}$. For fixed attention weights this is additive in the letters, so it cannot represent the indicator $[x_i = x_{i-N}]$, which is not additive (it is an XOR-like function of the two letters). Any dependence of the decision on equality must enter through a_{ij}, which depends on x_i and x_j jointly through the content term $c(x_i, x_j)$. □

Corollary 1. Write $r_i = \sum_j \bar{a}_j u_j + \sum_j (a_{ij} - \bar{a}_j) u_j$. The first term is a sum of functions of single letters, $\bar{a}_j u(x_j, j)$. Because the generator is invariant under relabelling the alphabet, each x_j is uniform on both match and non-match trials, so the first term has the same expectation on both. The second term is at most $u_{\max} \|a_i - \bar{a}\|_1$ in absolute value. Taking expectations on both trial types gives the bound. If attention does not depend on content, a_i is fixed for each i; with $\bar{a} = a_i$ the bound is zero, so the means are equal. The distributions need not be equal (joint letter statistics differ between trial types), so a threshold can retain a small sensitivity. □

Corollary 2. On the match trial the target key carries the query letter and receives $a_t = e^{c^+ + p(N)}/Z^{\mathrm{m}}$, where $c^+ = c(x, x)$. On the paired lure trial the lure key carries it and receives $a_\ell = e^{c^+ + p(N \pm 1)}/Z^{\mathrm{l}}$. Hence $a_\ell / a_t = e^{-\delta_P} Z^{\mathrm{m}}/Z^{\mathrm{l}}$. The two normalizers differ only in the terms of these two keys, so the ratio is near 1 when those terms are a small share of either; when the matching key dominates Z^{m} the ratio exceeds 1 and weakens the prediction. □

Proposition 4. The class means of r are $\pm\gamma\, w^\top e + b_\pm$ (plus a common constant), so their difference is $2\gamma\, w^\top e + \beta$, and the within-class variance is $\sigma_D^2 + \sigma_z^2$. (a) The content enters h linearly, so scaling every g_j by $1 - \alpha$ scales β by $1 - \alpha$ and σ_D^2 by $(1 - \alpha)^2$; γ, e and z do not involve distractor letters. With $\beta \le 0$ and a positive numerator, the numerator rises and the denominator falls as α grows. (b) For a uniformly random q-dimensional subspace with projector P_R, $\mathbb{E}[P_R] = (q/d) I$ by rotational symmetry, so $\mathbb{E}\|P_R g\|^2 = g^\top \mathbb{E}[P_R] g = (q/d)\|g\|^2$. (c) The head's distractor contribution is $\sum_{j \in D} a_j w^\top W_O v_j$, and $\sum_{j \in D} a_j \le \epsilon$ because the distractors lie off the target line. □

Proposition 2. With target logit s_t and K competitors $s_k \le s_t - \Delta$, $a_t = e^{s_t}/(e^{s_t} + \sum_k e^{s_k}) \ge 1/(1 + K e^{-\Delta})$. Requiring $a_t \ge 1 - \epsilon$ gives $K e^{-\Delta} \le \epsilon/(1 - \epsilon)$, so $\Delta \ge \log(K(1 - \epsilon)/\epsilon)$. For a lure ℓ with $x_\ell = x_t$, $c(x_i, x_\ell) = c(x_i, x_t)$ and $a_\ell / a_t = \exp(p(i - \ell) - p(i - t)) = e^{-\delta_P}$. □

Proposition 3 (sketch). With absolute positions P_j and a first-layer head whose query–key product is $\beta\, P_i^\top R P_j$ with R chosen so that $P_i^\top R P_j$ is maximized uniquely at $j = i - N$ with gap $g > 0$, the head places mass at least $1/(1 + L e^{-\beta g})$ on $i - N$, where $L \le i$ is the number of keys; its output, added to the residual stream, makes $e_{x_{i-N}}$ available at position i alongside e_{x_i}, and an MLP or a second attention layer can compute equality of the two letters. Rotary codes provide the same relative-offset selectivity. Under ALiBi, the bias $-\lambda(i - j)$ is monotone in distance, so one head's maximum over positions with equal content is always the most recent, and offset $N > 1$ cannot be singled out. □

B Methods

Minimal models. Attention-only transformers ($d_{\text{model}} = 512$ for the main one-layer configuration, 128 for variants), Adam with learning rate 10^{-4}, batch 32, fresh sequences each step, up to 30,000 steps with early stopping on validation loss (patience 2,000); 50 seeds for the main configuration (E1) and 10 for each variant. Variants: depth and residual stream, learned, sinusoidal, rotary, ALiBi and no position code, softmax, scalable softmax, sigmoid and top-k attention; the cued multi-N model

is a two-layer residual model with a cue token giving N. Recurrent baselines (LSTM, LRU, linear attention, selective SSM) use state sizes 32, 128 and 512.

LLM evaluation. Prompts start with a one-sentence instruction and "N = n", then three demonstration sequences with correct answers and the test sequence. Letters, " m" and " -" are single tokens in every tokenizer (checked per model). Models up to 1B run in float32 and larger models in bfloat16. The chat format sends one user turn per letter and one assistant turn per answer.

Head ranking (L3). Clean and corrupted prompts differ only in the letter at $i - N$, flipping the answer. For each head we patch its output from the clean run into the corrupted run at all positions and report the recovered fraction of the mean logit-difference gap (set to missing when the gap is below 0.1).

Locate analysis (L7). Attention from the query letter token to all tokens of a line is summed per head and averaged over the top-5 heads. The answer probe is a logistic regression on the top-5 heads' outputs at the query, trained on up to 4,000 correct items and applied to errors.

Probe (L8). Items with a same-letter item at $i - N$ (matches) or at a lure offset (lures); logistic regression with L_2 penalty on standardized features, five folds split by sequence.

Interventions (L5, B1). Sharpening multiplies a head's attention logits by $1/\tau$; ablation replaces a head's output by its mean over calibration items; random heads are drawn from the same layers. Suppression replaces h by $h - \alpha\,((h - \mu)U)U^\top$ at every residual site (input to the first layer and every layer's output) over the tokens of each non-target line, where U spans the 19 centred class means of the 20 letters at test-item letter tokens. The random control uses a random orthonormal U of the same rank.

Compute. Minimal models mostly on a CPU cluster (about 1,060 core-hours), with some sweeps on single H100 GPUs; LLM studies on single L40S, H100 or B200 GPUs per model and N. The whole project, including preliminary and failed runs, used about 51 GPU-hours.

Licenses. Pythia (Apache-2.0), GPT-2 (MIT), Qwen2.5 (Apache-2.0; Qwen license for some sizes).

C Additional results

Table 2: d' by model and N (line format, feedback condition), and false-alarm rates on N$\pm$1 lures versus non-lure items at N = 2.

Model	N=1	2	3	4	5	6	lure FA, N=2
Pythia-70m	1.25	1.52	1.40	0.03	0.45	-0.06	0.83 / 0.12
Pythia-160m	2.10	1.85	1.92	0.96	0.79	0.10	0.79 / 0.02
Pythia-410m	2.37	1.76	1.25	0.53	0.56	0.44	0.76 / 0.04
Pythia-1b	2.57	2.06	1.46	0.64	0.73	0.54	0.83 / 0.04
Pythia-1.4b	3.03	2.21	1.36	0.99	0.74	0.80	0.73 / 0.02
Pythia-2.8b	2.71	2.10	1.42	0.94	0.87	0.63	0.82 / 0.03
Pythia-6.9b	3.19	1.93	1.26	1.07	1.08	1.00	0.64 / 0.03
GPT-2	2.15	2.08	1.28	0.87	0.73	-0.02	0.96 / 0.08
GPT-2 medium	2.27	1.92	1.35	0.95	0.89	0.50	0.92 / 0.06
GPT-2 large	2.36	2.24	1.43	0.91	0.67	0.21	0.95 / 0.04
GPT-2 xl	2.34	1.48	0.93	-0.01	0.16	0.15	0.70 / 0.06
Qwen2.5-0.5B	2.58	1.64	1.04	0.64	0.52	0.52	0.81 / 0.06
Qwen2.5-1.5B	2.86	2.18	1.75	1.18	1.26	0.76	0.65 / 0.01
Qwen2.5-7B	3.17	1.53	1.44	0.88	0.61	0.79	0.64 / 0.06
Qwen2.5-14B	4.28	2.42	1.60	1.29	1.41	1.15	0.63 / 0.01

Post-hoc attention edits in the minimal model (E7). On 720 one-layer checkpoints, a fixed split of attention between i and $i - N$ gives $d' \approx 0$ at every N; keeping only the model's own mass on i and $i - N$ lowers d' from 2.70 to 0.63 at N = 1 and from 2.18 to 1.32 at N = 6, with false alarms rising from 0.03 to 0.67 at N = 1. Removing either the content or the position term of the logits gives $d' \approx 0$.

Table 3: Change in d' under each intervention; bold entries have 95% bootstrap intervals excluding zero. † degenerate baseline (excluded from summaries).

Model	N	base d'	sharpen	ablate top	ablate rand.	suppress	supp. rand.
Pythia-70m	2	1.52	**-0.11**	**-1.71**	**-1.08**	**+0.38**	**-0.30**
	3	1.40	**+0.22**	**-1.49**	**-1.28**	**+0.05**	**-0.21**
	4	0.03†	**+0.97**	**-0.17**	+0.04	**+0.82**	**+0.31**
Pythia-160m	2	1.85	**+0.14**	**-1.62**	+0.01	**+1.25**	**+0.29**
	3	1.92	**-0.17**	**-1.90**	**-0.13**	**+0.30**	+0.03
	4	0.96	**+0.30**	**-1.09**	**-0.20**	**-0.22**	+0.01
Pythia-410m	2	1.76	**+0.17**	**-1.36**	**-0.03**	**+0.61**	-0.01
	3	1.25	**-0.08**	**-1.18**	**+0.13**	**+0.60**	**-0.04**
	4	0.53	**+0.13**	**-0.39**	**+0.08**	**+0.70**	**+0.05**
Pythia-1b	2	2.06	**-0.04**	**-1.81**	**+0.13**	**+0.59**	**-0.05**
	3	1.46	**+0.11**	**-1.70**	**+0.34**	**+0.26**	**-0.12**
	4	0.64	**+0.09**	**-0.83**	**+0.29**	**+0.39**	**-0.09**
Pythia-1.4b	2	2.21	**+0.08**	**-1.61**	**-0.06**	**+0.95**	**-0.05**
	3	1.36	**+0.05**	**-0.93**	**+0.05**	**+0.46**	**+0.05**
	4	0.99	**+0.05**	**-0.66**	+0.02	**+0.44**	**-0.07**
Pythia-2.8b	2	2.10	**-0.05**	**-1.59**	**-0.12**	**+0.25**	**-0.03**
	3	1.42	**+0.12**	**-1.01**	**-0.03**	**+0.34**	-0.01
	4	0.94	**+0.12**	**-0.70**	**+0.02**	**+0.41**	**+0.05**
Pythia-6.9b	2	1.93	**+0.03**	**-1.02**	**-0.08**	**+0.94**	**-0.08**
	3	1.26	**+0.10**	**-0.34**	**+0.26**	**+1.47**	**+0.13**
	4	1.07	**+0.07**	**-0.55**	+0.03	**+1.13**	**+0.09**
GPT-2	2	2.08	**-0.30**	**-2.20**	**-0.09**	**+0.05**	**-0.22**
	3	1.28	-0.00	**-1.12**	**-0.09**	**+0.71**	**-0.07**
	4	0.87	**-0.02**	**-0.86**	**+0.07**	**+0.32**	**-0.12**
GPT-2 medium	2	1.92	+0.02	**-1.38**	**+0.14**	**+1.05**	+0.00
	3	1.35	**-0.06**	**-1.43**	**-0.10**	**+0.86**	**-0.08**
	4	0.95	**-0.07**	**-0.98**	**-0.18**	**+0.34**	**-0.12**
GPT-2 large	2	2.24	**-0.17**	**-1.16**	**-0.09**	**+0.68**	**+0.06**
	3	1.43	**+0.15**	**-0.89**	**+0.02**	**+0.77**	**-0.03**
	4	0.91	**+0.06**	**-0.50**	**-0.05**	**+0.55**	**-0.03**
GPT-2 xl	2	1.46	**+0.12**	**-1.24**	**-0.06**	**+0.52**	**+0.15**
	3	0.94	**+0.12**	**-0.88**	+0.01	**+0.81**	**-0.05**
	4	-0.02†	**+0.55**	**-0.16**	**+0.53**	**+0.91**	**+0.48**
Qwen2.5-0.5B	2	1.64	**-0.03**	**-0.87**	**+0.06**	**+0.49**	**+0.15**
	3	1.04	**+0.05**	**-0.65**	**-0.11**	**+0.67**	**-0.07**
	4	0.64	-0.01	**-0.38**	+0.02	**+0.60**	**-0.09**
Qwen2.5-1.5B	2	2.18	**+0.10**	**-0.76**	**+0.06**	**+0.89**	**-0.11**
	3	1.75	+0.01	**-0.93**	**-0.05**	**+1.15**	+0.01
	4	1.18	**-0.04**	**-0.78**	**-0.11**	**+0.28**	**-0.09**
Qwen2.5-7B	2	1.53	+0.01	**-0.27**	**+0.06**	**+0.41**	**+0.04**
	3	1.44	**-0.11**	**-0.29**	-0.02	**+0.99**	**+0.20**
	4	0.88	+0.00	+0.03	+0.00	**+0.63**	**+0.14**
Qwen2.5-14B	2	2.42	**+0.05**	**-0.48**	**-0.04**	**+0.46**	**-0.10**
	3	1.60	+0.01	**-0.17**	+0.01	**+1.78**	**-0.05**
	4	1.29	**+0.13**	**-0.21**	**+0.07**	**+0.76**	**+0.18**

Table 4: Locate (L7) and probe (L8) results per model.

		L7: share of misses			L8: match vs. lure		
Model	N	elsewhere	lost later	head wrong	model AUC	probe AUC	FA called lure
Pythia-70m	2	0.08	0.49	0.43	0.58	0.97	0.75
	3	0.01	0.42	0.57	0.57	0.91	0.61
Pythia-160m	2	0.26	0.34	0.40	0.63	0.98	0.82
	3	0.03	0.59	0.38	0.58	0.99	0.85
Pythia-410m	2	0.11	0.36	0.53	0.66	1.00	0.95
	3	0.06	0.40	0.54	0.58	0.98	0.84
Pythia-1b	2	0.11	0.32	0.57	0.68	1.00	0.95
	3	0.06	0.30	0.64	0.60	0.98	0.87
Pythia-1.4b	2	0.12	0.72	0.16	0.76	1.00	0.99
	3	0.06	0.38	0.56	0.62	0.99	0.92
Pythia-2.8b	2	0.04	0.36	0.60	0.71	1.00	1.00
	3	0.04	0.38	0.58	0.60	1.00	0.96
Pythia-6.9b	2	0.19	0.55	0.26	0.74	1.00	1.00
	3	0.08	0.38	0.54	0.60	1.00	0.98
GPT-2	2	0.16	0.65	0.19	0.49	0.97	0.84
	3	0.12	0.38	0.50	0.48	0.84	0.53
GPT-2 medium	2	0.18	0.53	0.29	0.48	0.99	0.87
	3	0.07	0.39	0.54	0.47	0.86	0.57
GPT-2 large	2	0.72	0.15	0.13	0.56	0.99	0.92
	3	0.54	0.20	0.26	0.50	0.88	0.59
GPT-2 xl	2	0.15	0.42	0.43	0.60	1.00	0.92
	3	0.10	0.28	0.62	0.49	0.89	0.69
Qwen2.5-0.5B	2	0.16	0.50	0.34	0.62	1.00	1.00
	3	0.09	0.23	0.67	0.53	0.99	0.91
Qwen2.5-1.5B	2	0.17	0.42	0.41	0.78	1.00	1.00
	3	0.04	0.39	0.57	0.67	0.99	0.86
Qwen2.5-7B	2	0.14	0.43	0.43	0.63	1.00	0.98
	3	0.45	0.19	0.36	0.60	0.98	0.88
Qwen2.5-14B	2	0.68	0.17	0.14	0.81	1.00	1.00
	3	0.51	0.13	0.36	0.64	1.00	0.95

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state the four claims (decomposition, causal contrast, where precision binds, architectures); each maps to Sections 5–7 and states its scope (15 base models up to 14B parameters, letter N-back).

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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